RiskScout

present

Invisible Data:
The Largest Problem Facing AI





PRESENTER:

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Today's Roadmap

- The Noisy Real World
- Factors to Consider
- Cybercrime
- Sports Marketing
- Financial Crime
- Lessons Learned
- Practical Solutions



The Noisy Real World

The Noisy Real World

- Real world doesn’t have to make sense
- ML invented to find patterns
- Found patterns aren’t required to exist
- Bad data introduces bad patterns

Factors to Consider

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- **Pareto Principle / Survivorship Bias**
  - 80% of data belongs to 20% of labels
- **Historical Bias**
  - Trends can change
- **Selection Bias**
  - Prove training data is representative of real data
- **Measurement Bias**
  - Are measurement techniques consistent across samples

Factors to Consider (cont.)

- **Reporting Bias**
  - Good news reported frequently, not bad
- **Confirmation Bias**
  - Nobody likes being proven wrong
- **Omission Bias**
  - Hiding information to mislead observers

Cybersecurity

Cybersecurity

- **Skilled Cybercriminals Leave Little Trace**
  - Better Tools, Techniques, and Experience
- **Most Cybercriminals Use Low-Effort Attacks**
  - Noisy, Easy-to-Spot
- **Cutting Edge Exploit Development**
  - Limited public data available
- **Patches Made Available**
  - Old exploits become less useful over time

Cybersecurity (cont.)

- **New detection tools built**
  - New data becomes available
- **New evasion techniques are refined**
  - Old techniques only work on certain subset of attacks
- **Victims under-report cyberattacks**
  - Shame
  - Reputational damage
  - Financial loss

Sports Marketing

Computer Vision in Sports Marketing

Sports Marketing

- **Most Placements Occur Infrequently**
  - Cameras focused on sports
  - No guarantee that placement will be close to action
- **80/20 Ratio in Placement Screen Time**
  - Bigger brands = More Budget
- **Teams Constantly Changing Partnerships**
  - Can change between seasons
  - Midseason changes (FTX Goes Under, OF & the Middle East)

Computer Vision in Sports Marketing (cont.)

- **Brands Optimizing Logo and Signage**
  - Can change strategy at any time
- **Environment**
  - Camera Hardware/Settings
  - Occlusion
  - Lighting
  - Atmospheric Effects

Financial Crime

Money Laundering

- **Differing Banking Cores**
  - Similar underlying data, but each one has its own quirks
- **Criminals Don’t Announce Crimes**
  - Want their money before regulators know what happened
- **Differing Hours and Internal Practices**
  - Which days are they open?
- **Differing Number of Branches**
  - Can customers access accounts at any branch?

Money Laundering (cont.)

- **Differing Customers by Region**
  - Lifestyle and Industry have Geographical Clustering
- **Transaction Size and Frequency Vary Widely**
  - Products, services, and brand affinity
- **Business vs. Personal Accounts**
  - Different needs and habits for each
- **Each FI has Limited Information**
  - One FI has limited info on activity outside their purview

Lessons Learned

Lessons Learned



Lessons Learned

- **Understanding the environment is critical**
  - Info available
  - Detecting shifts
- **Feature Engineering**
  - Not all columns are equally novel and valuable
  - Giving too much data kills performance

Lessons Learned (cont.)



Lessons Learned

- **Clearly Define Problem**
  - What's actually making things difficult?
  - Is there a deterministic solution?
- **Verify Data Reliability**
  - What are common issues?
  - Is there duplicated/missing data?
  - Detecting false positives and negatives?

Practical Solutions

Practical Solutions

- **Weighted Loss Functions**
  - Not always possible to do
  - Designed for class imbalances
- **Improved Data Collection**
  - Find under-reported data
  - Improve consistency
  - Fix mistakes

Practical Solutions (cont.)

- **Identify known gaps**
  - Available fields vs Relevant fields
  - Imputing unknown from known
- **Clean dataset**
  - Remove duplicates
  - Make available fields consistent
  - Images/Video needs same size

Conclusion

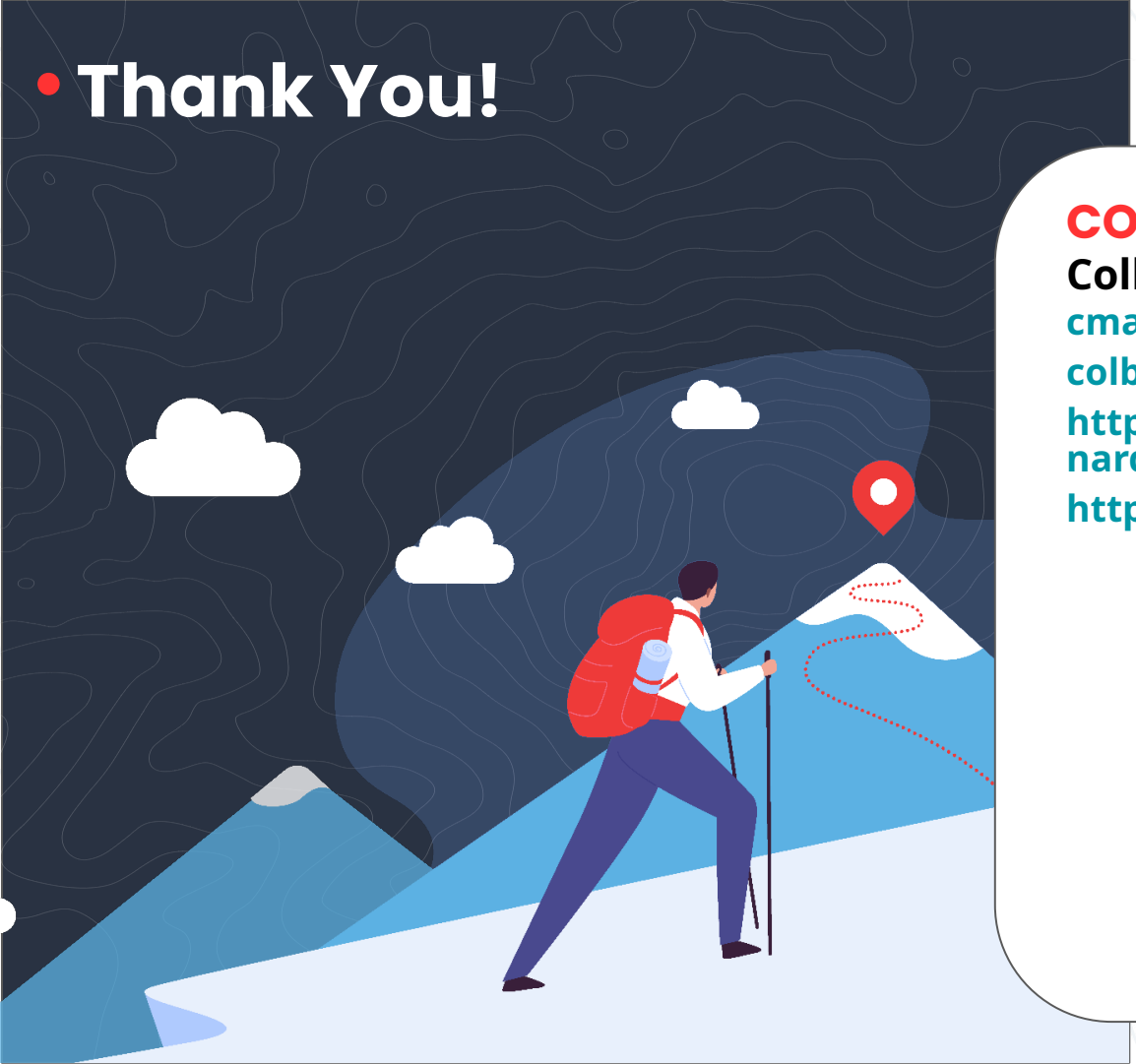
Conclusion

- **Remove Biases First**
  - ML useless with bad data
  - Get to know the data before starting
- **Identify Shifts Over Time**
  - Real world events impact raw data
  - Relationships between columns can change
  - Models are static
  - PSI vs CSI for automatic measurements

Conclusion

- **Impute Needed Info When Not Supplied**
  - Relationships with stable fields
  - Could have deterministic formula
  - Make sure policy for imputing data is clear

Thank You!



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